Chapter 5. From Representation to Prediction: Theorizing the AI Image

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In this chapter I will describe several characteristics of visual generative media at the current stage of its development that I believe are particularly significant or novel. Some of my arguments also apply to generative media in general, but mostly I focus on AI images. My approach to thinking about AI media is informed not only by my previous theoretical and artistic work with digital media but also practical experience of using popular AI image tools such as Midjourney almost daily for two years starting from August 2022.

The theories of AI images offered in the chapter synthesize ideas and perspectives from several fields: history of modern art, history of digital media, media theory, and software studies. I examine parallels between current AI practices and historical artistic movements. I also contextualize AI art within the broader history of media creation, examining how it builds upon and diverges from previous methods of image generation. My research methods involve comparing AI media with traditional artistic practices, conducting extensive experiments with AI image generation tools, and examining and testing the capabilities and limitations of these systems.

The Terms

Let’s begin by defining the terms. The terms generative media, synthetic media, AI media, generative AI, GenAI are all interchangeable. They refer to the process of creating new media artifacts with software tools that use certain types of artificial neural networks (e.g., AI models) trained on vast collections of media objects already in existence. The artifacts that these models can generate include images, animation, video, songs, music, text, music, 3D models and scenes, code, synthetic data, and

other types of media. While generation of new media objects received lots of public attention since 2022, today (middle of 2024) the more common use of these tools in Creative Industry is for media editing. For example, a writer can ask ChatGPT or another AI language bot to help editing an article or generating its abstract, while a photographer can use generative fill tool in Photoshop to replace any area of an image with another content that fits visually with the content outside of this area.

This chapter focuses on a particular type of generative media: images. Such images made with AI tools can be also referred by other terms such as generative images, synthetic images, AI images, and AI visuals. Note that the word “generative” itself can be used in different ways to refer to making cultural artifacts using any algorithmic processes (as opposed to only generative AI models) or more generally, any rule-based process that does not use computers. This is how the phrases generative art and generative design are typically used today in cultural discourses and popular media. In this chapter I am using generative in more restrictive way to designate AI models and GenAI apps for media generation and editing that use these methods.

’A I’ as a Cultural Perception

There is not one specific technology, or a single research project called ‘A I’ . However, we can follow how our cultural perception of this concept evolved over time and what it was referring to in each period. In the last fifty years, when an allegedly uniquely human ability or skill is being automated by means of computer technology, we refer to it as ‘A I’ . Yet, as soon as this automation is seamlessly and fully successful, we tend to stop referring to it as an ‘AI case’ . In other words, ‘A I’ refers to technologies and methodologies that automate human cognitive abilities and are starting to function but aren't quite there yet. ‘A I’ was already present in the earliest computer media tools. The first interactive drawing and design system, Ivan Sutherland’s Sketchpad (1961- 1962), had a feature that would automatically finish any rectangles or circles you started drawing. In other words, it knew what you were trying to make. In the very broad understanding just given, this was undoubtedly ‘A I’ already.

My first experience with a desktop paint program running on an Apple II was in 1984, and it was truly amazing to move your mouse and see simulated paint brushstrokes appear on the screen. However, today we no longer consider this to be ‘A I’ . Another

example would be the Photoshop function that automatically selects an outline of an object. This function was added many years ago – this, too, is ‘A I’ in the broad sense, yet nobody would refer to it as such today. The history of digital media systems and tools is full of such ‘AI moments’ – amazing at first, then taken for granted and forgotten as ‘A I’ after a while. (In academic studies of AI history, this phenomenon is referred to as the AI effect.) Thus, today creative AI refers only to recently developed methods where computers transform some inputs into new media outputs (e.g., text-to- image models) and specific techniques (e.g., certain types of AI models). However, we must remember that these methods are neither the first nor the last in the long history and future of simulating human art abilities or assisting humans in media creation. I expect that after a certain period, GenAI technology will be taken for granted, becoming ubiquitous and thus invisible – and some other cultural use of computers will come to be seen as 'AI. '

“Make it New” : AI and Modernism

After training on trillions of text pages or billions of images taken from the web, A I models can generate new texts and visuals on the level of highly competent professional writers, artists, photographers, or illustrators. These capacities of the A I models are distributed over trillions of connections between billions of artificial neurons rather than determined by standard algorithms. In other words, we developed a technology that, in terms of complexity, is extremely similar to the human brain. We don't fully grasp how our AI technology works, just as we don't fully comprehend human intellect and creativity.

The current generation of generative AI image models and tools, such as Midjourney and Stable Diffusion, have been trained on very large and diverse datasets containing hundreds of millions or billions of images and their text descriptions. It is, however, equally interesting to limit the training data set to a specific area within the larger space of human cultural history, or to a specific set of artists from a specific historical period. Unsupervised by Refik Anadol Studio (2022) is AI art project that exemplifies these possibilities. The project uses AI models trained on the image dataset oftens of

thousands of artworks from the MoMA collection. This collection, in my opinion, is one of the best representations of the most creative and experimental period in human visual history - hundred years of modern art (1870 - 1970) - as well as many important

examples of artistic explorations in the subsequent decades. It captures modernist artists ' feverish and relentless experiments to create new visual and communication languages and "make it new."



Chapter_05_figure_01. Refik Anadol Studio, Unsupervised, 2022. Selected frames from the animation.

On the surface, the logic of modernism appears to be diametrically opposed to the process of training generative AI systems. Modern artists desired to depart from classical art and its defining characteristics such as visual symmetry, hierarchical compositions, and narrative content. In other words, their art was founded on a fundamental rejection of everything that had come before it (at least in theory, as expressed in their manifestos). AI models are trained in the opposite manner, by learning from historical culture and art created up to now. AI model is analogous to a very conservative artist studying in the "meta" museum without walls that houses historical art.

But we all know that art theory and art practice are not the same thing. Modern artists did not completely reject the past and everything that came before them. Instead,

modern art developed by reinterpreting and copying images and forms from old art traditions, such as Japanese prints (van Gogh), African sculpture (Picasso), and Russian icons (Malevich). Thus, the artists only rejected the dominant high art paradigms of the time, realistic and salon art, but not the rest of human art history. In other words, it was deeply historic ist: rather than inventing everything from scratch, it innovated by adapting certain older aesthetics to contemporary art contexts. (In the case of geometric abstract art created in 1910s, these artists used images that were already widely used in experimental psychology to study human visual sensation and perception.1

When it comes to artistic AI, we should not be blinded by how these systems are trained. Yes, AI models are trained on previously created human art and culture artifacts. However, their newly

generated outputs are not mechanical replicas or simulations of what has already been created. In my opinion, these are frequently genuinely new cultural artifacts with previously unseen content, aesthetics, or styles.

Of course, simply being novel does not automatically make something culturally or socially interesting or significant. Indeed, many definitions of "creativity" agree on this point: it is the creation of something that is both original and worthwhile or useful.

However, estimating what percentage of all novel artifacts produced by generative A I are also useful and/or meaningful for a larger culture is not a feasible project at this time. For one thing, I am not aware of any systematic effort to use such systems to "fill in," so to speak, a massive matrix of all content and aesthetic possibilities by providing millions of specifically designed prompts. Instead, it is likely that, as in every other area of popular culture, only a small number of possibilities are realized over and over by millions of users, leaving a long tail of other possibilities unrealized. So, if only a tiny fraction of the vast universe of potential AI artifacts is being realized in practice, we can't make broad statements about the originality or utility of the rest of the universe.

Generative Media and Database Art

Some AI artists such Anna R idler2, Sarah Meyohas3 and Refik Anadol4 utilized in their works AI models trained on specific datasets. Many other artists, designers, architects, and technologists use models released by other companies or research institutions that

were already trained on very large datasets (e.g., Stable Diffusion), and then fine tune them on their own data.

For example, artist Lev Pereulkov5 fine-tuned the Stable Diffusion model 2.1 using 40 paintings by well-known “ non-conformist” artists who worked in USSR starting in the 1960s (Erik Bulatov, Ilya Kaba kov, and others). Pereulkov 's image series Artificial Experiments 1–10 (2023) created with this customized AI model, is an original piece of art that captures the artistic characteristics of these artists as well as their unique surreal and ludicrous semantics without repeating closely any of their existing works.6 Instead, their "DNAs" captured by the model enable new meanings and visual concepts.



Chapter_05_figure_02. Lev Pereulkov, Artificial Experiments 1–10, 2023. Three images from the series of 10 shared on Instagram.

Most of the millions of everyday people and creative professionals who employ generative media tools use them as is, and don't fine them further. This may change in the future as the techniques fine tuning AI models using our own data may become easier to use and wider available. But regardless of these specifics, all newly created cultural artifacts produced by AI models have a common logic.

Unlike traditional drawings, sculptures, and paintings, generative media artifacts are not created from scratch. They are also not the result of capturing some sort of sensory phenomenon, such as photos, videos, or sound recordings. They are instead built from

large archives of already existing media artifacts. This generative mechanism links generative media to earlier art genres and processes.

We can compare it to film editing, which first appears around 1898, or even earlier composite photography, which was popular in the nineteenth century. We can also consider specific artworks that are especially relevant, such as experimental collage film A Movie (Bruce Conner, 1958) or many Nam June Park installations that feature bits of television programs playing across hundreds of TV monitors.

Seeing projects like Unsupervised or artificial experiments 1-10 in the context of this media creation method and its historical variations will help us understand this and many other AI artworks as art objects engaged in dialogues with art from the past, rather than as purely technological novelties or works of entertainment.

I see many relevant moments and periods when I scan the history of art, visual culture, and media for other prominent uses of this procedure. They are relevant to the current generative media not only because artists working at these times used the procedure, but also because the reason for this use was consistent in all cases. A new accumulation and accessibility of masses of cultural artifacts led artists to create new forms of art driven from these accumulations. Let me describe a few of these examples.

Net and digital artists created a number of works in the late 1990s and early 2000s in response to the new and rapidly expanding universe of the world wide web. Health Bunting 's _readme (1998), for example, is a web page containing the text of an article about the artist, with each word linked to an existing web domain corresponding to that word. Mark Napier 's Shredder 1.0 (also 1998) presents a dynamic montage of elements that comprise numerous websites—images, texts, HTML code, and links.

Going further back in time, we find a broad cultural paradigm that was also a reaction to the accumulation of historical art and culture artifacts in easily accessible media collections. This is paradigm is known as "post-modernism." Post-modern artists and designers frequently used bricolage and created works consisting of quotations and references to art from the past, rejecting modernism 's focus on novelty and breaking with the past.

While there are many possible explanations for the emergence of the post-modern paradigm in the 1960s and 1980s, one is relevant to our discussion. The accumulation

of earlier art and media artifacts in structured and accessible collections such as slides libraries, film archives, art history textbooks with many photos of the artworks, and other formats - where different

historical periods, movements, and creators were positioned together - inspired artists to begin creating bricolages from such references as well as extensively quoting them.



Chapter_05_figure_03. Photomontage by John Heartfield, 1919.

What about artistic modernism of the 1910s and 1920s? While the overall emphasis was on originality and novelty, one of the procedures it developed in search of novelty was direct quotations from the vast universe of contemporary visual media that was rapidly expanding at the time. Large headings, for example, and the inclusion of photos

and maps made newspapers more visually impactful; new visually oriented magazines, such as Vogue and Times, were also launched in 1913 and 1923, respectively; and of course, a new medium of cinema continued to develop.

In response to this visual intensification of mass culture, in the early 1910s Georges Braque and Pablo Picasso began incorporating actual newspaper, poster, wallpaper, and fabric fragments into their paintings. A few years later, John Heartfield, George Grosz, Hannah Hoch, Aleksandr Rodchenko, and a handful of other artists began to develop photo-collage techniques. Photo-collage became another method of creating new media artifacts from existing mass media images.

Contemporary artworks that employ AI models trained on cultural databases, such as Unsupervised or artificial experiments 1-10, continue a long tradition of creating new art from accumulations of images and other media. In this way, these works of art keep opening up new possibilities for art and its techniques, particularly those of what I referred to already in 1998 as "database art."7 The introduction of new methods for reading cultural databases and creating new narratives from them is part of this expansion.

Thus, Unsupervised neither creates collages from existing images, as did modernist artists of the 1920s, nor quotes them extensively, as did postmodern artists of the 1980s. Instead, the members of Anadol Studio train a neural network to extract patterns from tens of thousands of MoMA 's artworks. The trained net then generates new images that share the same patterns but don't look like any specific paintings. Throughout the course of the animation, we travel through the space of these patterns (e.g., "latent space"), exploring various regions of the universe of contemporary art.8

Pereulkov 's Artificial Experiments 1–10 use a different technique to generate new images from an existing image database. He chose only forty paintings by artists who share key characteristics. They developed their oppositional art in late communist society (USSR, 1960s-1980s). They also lived in the same visual culture. In my memories, this society was dominated by two colors: grey (representing the monotony of urban life) and the red of propaganda. In addition, Pereulkov chose paintings that share something else: "I chose, as a rule, paintings that conceptually relate in some way to the canvas - or to the space on it. I obtained the photo of a painting New Accordion from Kaba kov, which features paper applications on top of the canvas."9 Pereulkov also crafted custom text descriptions of

each painting used for fine-tuning the Stable Diffusion model. To teach the model the specific visual languages of the chosen artists, he added terms such as "thick strokes," "red lighting," "blue background," and "flat circles" to these descriptions.

Clearly, each of these steps represents a conceptual and aesthetic decision. In other words, the key to the success of Artificial Experiments 1–10 is the creation of such a database. This work demonstrates how fine-tuning an existing neural network that was trained on billions of image and text pairs (such as Stable Diffusion) can make this network follow artists ' ideas; the biases and noise of such a massive network can be overcome and minimized, and do not need to dominate our own imagination.

From Representation to Prediction

Historically, humans created images of existing or imagined scenes using a number of methods, from manual drawing to 3D CG (see below for explanation of the methods). With AI generative media, a fundamentally new method emerges. Computers analyze patterns in large datasets of existing media. Using these learned patterns, they can then create new, previously unseen still and moving images that exhibit similar characteristics. This process forms the core of Generative AI technology.

One can certainly propose different historical paths leading to visual generative media today or divide one historical timeline into different stages. Here is one such possible trajectory:

1. Creating representations manually (e.g. drawing with variety of instruments, carving, etc.). More mechanical stages and parts were sometimes carried out by human assistants typically training in their teacher’s studio – so there is already some delegation of functions. 2. Creating manually but using assistive devices (e.g. perspective machines, camera lucida). From hands to hands + device. Now some functions are delegated to mechanical and optical devices. 3. Photography, x-ray, video, volumetric capture, remote sensing, photogrammetry. From using hands to recording information using machines. From human assistants to machine assistants.

4. 3D CG. You define a 3d model in a computer and use algorithms that simulate effects of light sources, shadows, fog, transparency, translucency, natural textures, depth of field, motion blur, etc.

From recording to simulation. 5. Generative AI. Using media datasets to predict still and moving images. From simulation to prediction.

“ Prediction” is the actual term often used by AI researchers in their publications describing visual generative media methods. So, while this term can be used figuratively and evocatively, this is also what happens scientifically when you use image generative tools. When working with a text-to-image AI-model, the artificial neural network attempts to predict the images that correspond best to your text input. I am certainly not suggesting that using all other already accepted terms such as ‘generative media’ is inappropriate. But if we want to better understand the difference between AI visual media synthesis methods and other representational methods developed in human history, employing the concept of ‘prediction’ and thus referring to these AI systems as ‘predictive media’ captures this difference well.

Media Translations

There are several methods for creating ‘AI media’ . One method transforms human media input while retaining the same media type. Text entered by the user, for example, can be summarized, rewritten, expanded, and so on. The output, like the input, is a text. Alternatively, in the image-to-image generation method, one or more input images are used to generate new images.

However, there is another path that is equally intriguing from the historical and theoretical perspectives. ‘AI media’ can be created by automatically ‘translating’ content between media types. This is what happens, for example, when you are using Midjo iurney, Stable Diffusion or other AI image generator service and enter a text prompt, and AI generates one or more images in response. Text is ‘translated’ into an image.

Because this is not a literal one-to-one translation, I put the word ‘translation’ in quotes. Instead, input from one medium instructs AI model to predict the appropriate output from another. Such input can also be said to be ‘ mapped’ to some outputs in other media. Text is mapped into new styles of text, images, animation, video, 3D models,

and music. The video is converted into 3D models or animation. Images are ‘translated’ into text, and so on. Text-to-image method translation is currently more advanced than others, but various forms will catch up eventually.

Translation (or mapping) between one media and another is not a new concept. Such translations were done manually throughout human history, often with artistic intent. Novels have been adapted into plays and films, comic books have been adapted into television series, a fictional or non-fictional text was illustrated with images, etc. Each of these translations was a deliberate cultural act requiring professional skills and knowledge of the appropriate media. Some of these translations can now be performed automatically on a massive scale thanks to AI models, becoming a new means of communication and culture creation. Of course, artistic adaptation of a novel into a film by a human team and automatic generation of visuals from novel text by a net is not the same thing, but for many more simple cases automatic media translation can work well. What was once a s killed artistic act is now a technological capability available to everyone. We can be sad about everything that might be lost because of the automation – and democratization – of this critical cultural operation: skills,

something one might call ‘deep artistic originality’ or ‘deep creativity’, and so on. However, any such loss may be only temporary if the abilities of ‘culture A I’ are, for example, even further improved to generate more original content and understand context better.

Because most people in our society can read and write in at least one language, text-to- another media methods are currently the most popular. They include text-to-image, text-to-animation, text-to-3D, and text-to-music models. These AI tools can be used by anyone who can write, or by using readily available translation software to create a prompt in any of the language these tools understand best at a given point. However, other media mappings can be equally interesting for professional creators. Throughout the course of human cultural history, various translations between media types have attracted attention. They include translations between video and music done by VJs in clubs; long literary narratives turned into movies and television series; texts illustrated with images in various media such as engravings; numbers turned into images (digital art); texts describing paintings (ekphrasis tradition, which began in Ancient Greece), mappings between sounds and colors (especially popular in modernist art); etc.

The continued development of AI models for mappings between all types of media, without privileging text, has the potential to be extremely fruitful, and I hope that more

tools will be able to accomplish this. Such tools will be very useful both to professional artists and other creators a like. However, being an artist myself, I am not claiming that future ‘culture A I’ will be able to match, for example, innovative interpretations of Hamlet by avant-garde theatre directors such as Peter Brook or astonishing abstract films by Oscar Fishinger that explored musical and visual correspondences. It is sufficient that new media mapping AI tools stimulate our imagination, provide us with new ideas, and enable us to explore numerous variations of specific designs.

The Stereotypical and the Unique

Both the modern human creation process and the predictive AI generative media process seem to function similarly. AI model is trained using unstructured collections of cultural content, such as billions of images and their descriptions or trillions of web and book pages. The neural net learns associations between these artifacts’ constituent parts (such as which words frequently appear next to one another) as well as their common patterns and structures. The trained net then uses these structures, patterns, and ‘culture atoms’ to create new artifacts when we ask it to. Depending on what we ask for, these AI-created artifacts might closely resemble what already exists or they might not.

Similarly, our life is an ongoing process of both supervised and unsupervised cultural training. We take art and art history courses, view websites, videos, magazines, and exhibition catalogs, visit museums, and travel in order to absorb new cultural information. And when we ‘prompt’ ourselves to make some new cultural artifacts, our own biological neural and networks (infinitely more complex than any A I nets to date) generate such artifacts based on what we’ve learned so far: general patterns we’ve observed, templates for making particular things, and often concrete parts of existing artifacts. In other words, our creations may contain both exact replicas of previously observed artifacts and new things that we represent using templates we have learned, such as golden ratio or use of complementary colors.

AI tools for image generation frequently have a default "house style." (This is the actual term used by MidJourney developers.) If one does not specify a style explicitly, the A I tool will generate images using its default aesthetic. Each of the popular AI image tools and AI bots that can also generate images (ChatGPT, Anthropic 's Claude, Google Gemini, Microsoft Copilot, etc.) has its own default style; these styles can change from version to version.



Chapter_05_figure_04. Examples generated in Midjourney version 4 using the short text prompt “morning sky.”

To steer away from this default, you need to add terms to your prompts specifying a description of the medium, the kind of lighting, the colors and shading, or a phrase like “in the style of” followed by the name of a well-known artist, illustrator, photographer, fashion designer, or architect. Here are two examples of such prompts from my own artistic work, and the images that Midjourney generated from these prompts (2022- 2023). The terms used to define particular style characteristics are highlighted in italics.



Chapter_05_figure_05. Lev Manovich, an image generated with Midjourney version 3 (2022) using the following prompt: ”giant future 1965 modern airport in Siberia made from water and ice, painted on large wood panel by Hieronymus Bosch, bright pastel colors with white highlights, 23f lens, very detailed --ar 4:3 --s 1250 —test"



Chapter_05_figure_06. Lev Manovich, an image generated with Midjourney version 4 (2023) using the following prompt: “ Photo of two Russian high-school students, clear skin, very soft studio light, 50mm lens, monochrome, silver tones, high quality, ultra realistic --v 4 --q 2.” This image also illustrates the point I am making later in the chapter: “AI frequently generates new media artifacts that are more stereotypical or idealized than what we intended.”

Because it can simulate many thousands of already-existing aesthetics and styles and interpolate between them to create new hybrids, GenAI is more capable than any single human creator in this regard. However, at least so far, skilled and highly experienced human creators also have a significant advantage. Both humans and artificial intelligence are capable of imagining and representing nonexistent and existing objects and scenes a like. Yet, unlike AI image generators, human-made images can include unique content, unique details, and unique aesthetics that is currently beyond the capabilities of AI. In other words, at least as of now, highly skilled human

image makers can create all kinds of images that currently even the best AI models cannot generate as well.

"Unique" in this context means that this particular content, details, or visual aesthetics were "never" realized before. Why did I put the word "never" in quotes? Because no single cultural artifact in human history is 100% unique - we can always find other examples with at least some of the same (or very similar) details. Therefore, if we want to be more precise, we can say the following: a "unique cultural artifact" means that enough of its details are original, making it sufficiently distinct to appear novel and different from everything else.

AI image tools are much better at generating visuals of popular subjects and aesthetics than unfamiliar or rare ones. This reflects the abundance of these subjects on the web used as training data in the current stage of AI development. So how do these tools behave when you ask them to make something sufficiently unique – i.e., something nonexistent, rare, or unfamiliar to them? Based on my extensive use of the best A I image tools available during 2022-2024, I found that they react to such requests in one or both of these ways:

1) The quality of generated results becomes lower. Aesthetics are inferior, mistakes appear, and the generated scene is not coherent.

2) The tool substitutes the content and aesthetics you want with more commonplace alternatives. You get a coherent scene, figures and faces with correct anatomy, and interesting composition - but the image is composed of more generic (cliché) elements. The aesthetic parameters are shifted towards more common values.

In short, AI models either produces something different from what we ask for, or they may construct desired content from more stereotypical details.10

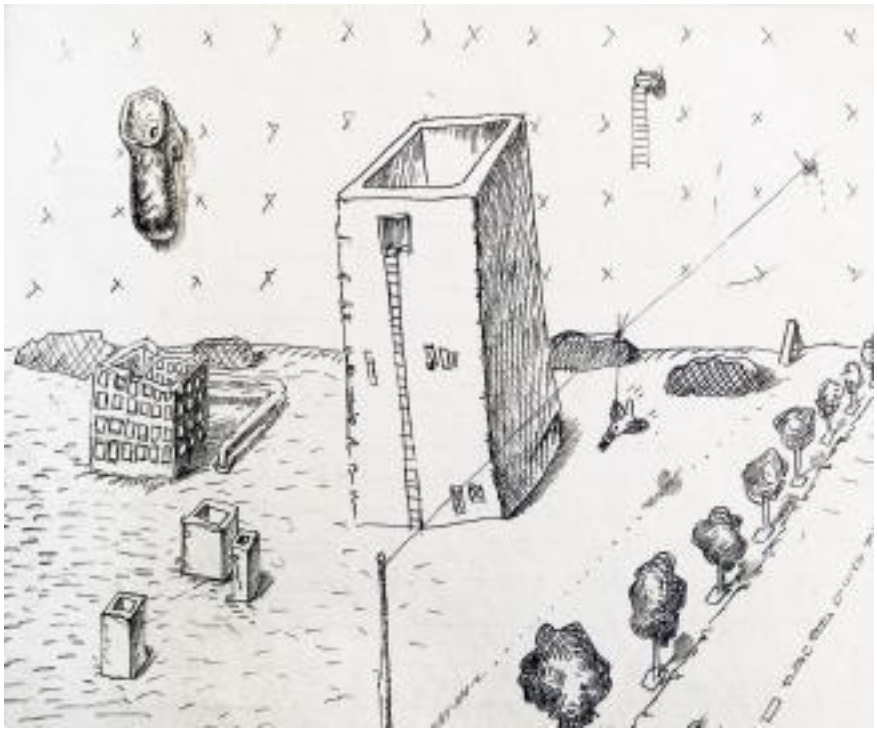
What is the cause of this aesthetic and content gap between human and artificial image making abilities? ‘Cultural atoms’, structures, and patterns in the training data that occur most frequently are very successfully learned during the process of training an artificial neural network. In the ‘ mind’ of a neural net, they gain more importance. On the other hand, details and structures that are rare in the training data or may only appear once are hardly learned or not even parsed at all. They do not enter the

artificial culture universe learned by AI. Consequently, when we ask AIto synthesize them, it is unable to do so.

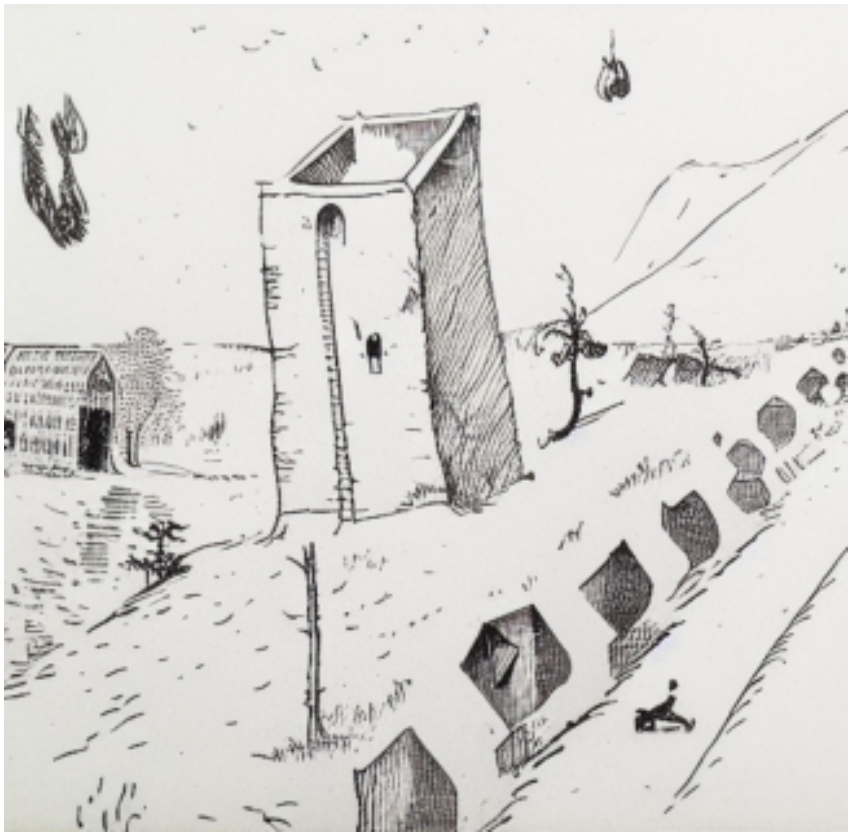
And this is why text-to-image AI tools such as Midjourney, Stable Diffusion, Leonardo.ai, or RunwayML are not currently able to generate perfectly new images in the style of my pen on paper

drawings (see Figure 7), expand my drawings by adding newly generated parts, or replace specific portions of my drawings with new content drawn perfectly in my style (e.g, they can't perform useful outpainting or inpainting on the digital photos of my drawings.)

Instead, these AI tools generate more generic objects than what I frequently draw or they produce images that are merely ambiguous yet uninteresting. (See Figure 8)



Chapter_05_figure_07. Lev Manovich, untitled drawing, pen on paper, 1981.



Chapter_05_figure_08. One of my attempts to generate a new version of the image from Figure 7 with Stable Diffusion AI, Fall 2022.

I am certainly not claiming that the style and the world shown in my drawings is completely unique. They are also a result of specific cultural encounters I had, things I observed, and things I noticed. But because they are uncommon (and thus unpredictable), AI finds it difficult to simulate them, at least without additional training using my drawings.

Here we encounter what I see as the key obstacle creators face when using generative AI generative tools - and this holds for both AI images and all other types of media:

Generative AI frequently generates media that are more stereotypical or idealized than what we intend.

This can affect any image dimensions - elements of content, lighting, crosshatching, atmosphere, spatial structure, and details of 3D shapes, among others. Occasionally it is immediately apparent, in which case you can either attempt to correct it or disregard the results. Very often, however, such ‘substitutions’ are so subtle that we cannot detect them without extensive observation or, in some cases, the use of a computer to quantitatively analyze numerous images.

In other words, new AI generative media models, much like the discipline of statistics since its inception in the 18th century and the field of data science since the end of the 2010s, deal well with frequently occurring items and patterns in the data but do not

know what to do with the infrequent and uncommon. We can hope that AI researchers will be able to solve this problem in the future, but it seems so fundamental that we should not anticipate a solution immediately.

Subject and Style

In the arts, the relationship between content and form has been extensively discussed and theorized. This brief section does not attempt to engage in all of these debates or to initiate discussions with all relevant theories. Instead, I would like to consider how these concepts play out in A I's 'generative culture' . However, instead of using content and form, I'll use a different pair of terms that are more common in AI research publications and online conversations between users: subject and style.

At first glance, AI media tools appear capable of clearly distinguishing between the subject and style of any given representation. In text-to-image models, for instance, you can generate countless images of the same subject. Adding the names of specific artists, media, materials, and art historical periods is all that is required for the same subject to be represented differently to match these references. Photoshop filters began to separate subject and style as soon in the 1990s, but AI generative media tools are more capable. For instance, if you specify "oil painting" in your prompt, simulated brushstrokes will vary in size and direction across a generated image based on the objects depicted. AI media tools appear to ' understand' the semantics of the representation as opposed to earlier filters that simply applied the same transformation to each image region regardless of its content. For instance, when I used "a painting by Malevich" and "a painting by Bosch" in the same prompt, Midjourney generated an image of space that contained Malevich-like abstract shapes as well as many small human and animal figures like in popular Bosch paintings that were properly scaled for perspective.



Chapter_05_figure_09. Lev Manovich, an image generated in Midjourney using prompt "painting by Malevich and Bosch," Fall 2022.

AI tools routinely add content to an image that I did not specify in my text prompt in addition to representing what I requested. This frequently occurs when the prompt includes “in the style of” or “by” followed by the name of a renowned visual artist or photographer. In one experiment, I used the same prompt with the Midjourney A I image tool 148 times, each time adding the name of a different photographer. The subject in the prompt remained mostly the same – an empty landscape with some buildings, a road, and electric poles with wires stretching into the horizon. Sometimes adding a photographer’s name had no effect on the elements of a generated image that fit our intuitive concept of style, such as contrast, perspective, and atmosphere. But

every now and again, Midjourney also modified the image content. For example, when well-known works by a particular photographer feature human figures in specific poses, the tool would occasionally add such figures to my photographs. (Like Malevich and Bosch, they were transformed to fit the spatial composition of the landscape rather than mechanically duplicated.) Midjourney has also sometimes changed the content of my image to correspond to a historical period when a well-known photographer created his most well-known photographs.

According to my observations, when we ask Midjourney or a similar tool to create an image in the style of a specific artist, and the subject we describe in the prompt is related to the artist’s typical subjects, the results can be very successful. However, when the subject of our prompt and the imagery of this artist are very different, ‘ rendering’ the subject in this style frequently fails.



Chapter_05_figure_10. Using prompt “by Caspar David Friedrich --v 5” in Midjourney generates images that capture the artist’s style sufficiently well. Image source: https://www.midlibrary.io/styles/caspar-david-friedrich.



Chapter_05_figure_11. Images generated in Midjourney v5 using a prompt: “decaying peonies by Caspar David Friedrich --v 5.” Images source: https://www.midlibrary.io/styles/caspar-david-friedrich.

Using prompt “decaying peonies by Caspar David Friedrich” in Midjourney generates images that simulate important features of artist’s style such as combinations of cool colors and dramatic atmosphere. But in other ways, generated images depart significantly from the artist’s style. The types of lines, rendering of details, and symmetrical compositions in these AI images would never appear in actual Friedrich’s paintings. AI can also often insert some generic looking objects, such as the rock formations in the upper right corner of first image.

To summarize, to successfully simulate a given visual style using current AI tools, you may need to change the content you intended to represent - or to accept the fact that AI will insert some details you don’t want. In other words, not every “subject” can be rendered successfully in any “style.” And what exactly AI can learn also varies from case to case: it can often successfully learn some features of artist’s style but not others.

In fact, after using AI image tools for two years and analyzing endless images generated by others, I realized that the popular idea that GenAI can perfect extract a “style” from a collection of aesthetically similar images is not correct. Instead, it appears that in the process of training, visual AI models learn associations between visual characteristics of

the works of a given artist (I.e., what we often call a visual language) and the content of their artworks (unless given artworks are fully abstract.) This puts into question the whole idea that style and content can be neatly separated.

These observations, I believe, complicate the binary opposition between the concepts of ‘content’ and ‘style’ . For some artists, AI can extract at least some aspects of their style from examples of their work and then apply them to different types of content. Sometimes more aspects of a given artist style can be learned and used for new generation, and sometimes less. But for other artists, it seems, their style and content cannot be separated.

For me, these kinds of observations and reflections are one of the most important reasons for using new media technologies like AI generative media and learning how they work. Of course, as a practicing artist and art theorist, I had been thinking about the relationships between subject and style (or content and form) for a long time—but being able to conduct systematic experiments like the one I described can lead to new ideas and allow us to look back at cultural history and our familiar concepts and theories of art and culture in new ways.

Notes

1 For the detailed analysis of these relations between modern art and experimental psychology, see Vitz, P., & Glimcher, A. (1983). Modern art and modern science: The parallel analysis of vision, https://muse.jhu.edu/pub/1/article/403181/pdf. 2 R idler, A. (n.d.). Anna Ridler. Retrieved September 28, 2024, from https://annaridler.com/. 3 Sarah Meyohas. AI Artists. Retrieved September 28, 2024, from https://aiartists.org/sarah-meyohas. 4 Anadol, R. RefikAnadol. Retrieved September 28, 2024, from https://refikanadol.com/. 5 Pereulye. Instagram. Retrieved September 28, 2024, from https://www.instagram.com/pereulye/. 6 Pereulkov. (2023, January 16). Artificial experiments 1–10. Instagram. https://www.instagram.com/p/CnezVZ9KHMV. 7 See my article Database as a symbolic form. (1998). https://manovich.net/index.php/projects/database-as-a-symbolic-form. 8 For a more details about GAN net training methods used by Refik Anadol Studio, see Creating art with generative adversarial network: Refik Anadol's Walt Disney Concert Hall Dreams. (2022). https://medium.com/@ymingcarina/creating-art-with-generative-adversarial-network-refik-anadols-wdc h-dreams-159a6eac762d. 9 My personal communication with Pereulkov, 04/16/2023. 10 For example, in 2022-2023 AI models often struggled generating realistic hands. By 2024 this has been solved but as I am writing this, AI still have difficulty generating very complex compositions with dozens of fully realistic human figures and faces shown in detail—something that skilled human artists were able to do at least since the Renaissance.